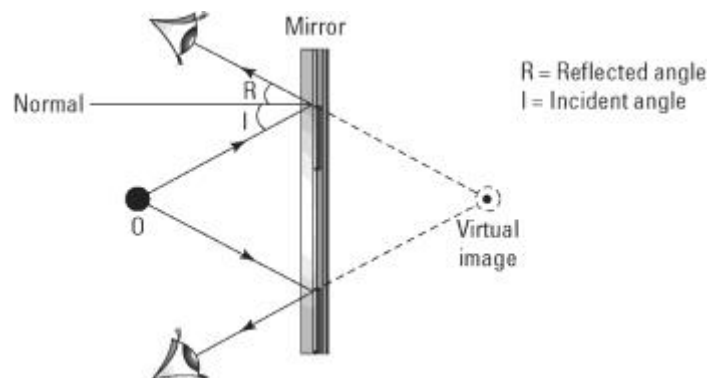


# Reflection and scattering of light

## Answers

### Locating the virtual image

1. An object (O) is observed by looking at the rays of light reflecting off a plane mirror. The mirror consists of a plate of glass with a silver back surface mounted on a sheet of wood.



- (a) Use a ruler and pencil to extend each reflected ray (using dotted lines) until they intersect behind the mirror. Label the point of intersection as 'virtual image'.
- (b) Complete the sentence:  
The object in front of a reflecting surface and a virtual image are the same distance from a mirror.
- (c) Draw the *normal* to the mirror surface and measure the incident and reflected angles for the ray above the object, O. From above diagram  $R = I$

### Light and air

2. The diagram below shows the collision of light rays with particles of air in the atmosphere. Use the letters X, Y and Z to identify the following rays.

Scattered: .....Z.....

Incident: .....X.....

Transmitted: .....Y.....

